

Ivan Reshetnikov

Senior ML / AI Engineer with 10+ years in software engineering and 5+ years in applied Data Science.

I build production ML systems for real business problems: matching, ranking, duplicate detection, pricing, anomaly detection, forecasting, and LLM/RAG-assisted data workflows. I usually work end-to-end, from understanding the problem and defining metrics to building, deploying, and improving the system after release.

GitHub: <https://github.com/resivalex>

Senior ML Engineer

Maksmart

<https://maksmart.ru>

B2B marketplace

1 year 10 months

August 2024 - Present

Building production ML systems for a B2B marketplace: product matching, duplicate detection, pricing, anomaly detection, catalog quality, delivery forecasting, and LLM/RAG-assisted data workflows.

Selected outcomes:

- Shipped 8 ML models to production across matching, pricing, catalog quality, and forecasting.
- Built a product matching system that reduced unmatched supplier offers by 2x.
- Developed duplicate scoring for implicit product duplicates, reducing duplicates by 60% at 95% precision.
- Built price regression with confidence estimates to support supplier offer moderation.
- Implemented price anomaly detection to identify unit-of-measure issues, data-entry errors, and suspicious supplier prices.
- Implemented last-mile delivery time forecasting with under 15% error.
- Designed LLM/RAG-assisted workflows for product type discovery, catalog maintenance, and data labeling.
- Built data processing and data quality pipelines, reducing memory consumption by 5x.

Technologies: Python, PyTorch, scikit-learn, LightGBM, fastText, AutoML, Active Learning, LLM/RAG, embeddings, FastAPI, Docker, PostgreSQL, ClickHouse

Data Scientist

Architech NYC

<https://architech.nyc>

Business services

2 years 4 months

April 2022 - July 2024

Built production ML, analytics, and automation tools for transportation and business operations: rate calculation, document validation, BI reporting, and email workflow automation.

Selected outcomes:

- Built a container transportation Rate Calculator with 11% RMSPE, monitoring, and regular retraining.
- Developed an OCR-based document validation service with 95% precision, reducing dispatcher workload.
- Built BI dashboards for transportation operations, improving visibility and reducing manual reporting.
- Implemented an AI-assisted email handling system using ChatGPT API and business data integrations to reduce response times.
- Investigated data consistency issues and prepared ad-hoc analyses for current business needs.

Technologies: Python, LightGBM, PyTorch, Streamlit, FastAPI, PostgreSQL, Docker, Apache Superset, ChatGPT API, pytest

Machine Learning Engineer

Sape

<https://sape.ru>

Backlinks purchase service

1 year 4 months

December 2020 - March 2022

Applied ML to SEO and search products operating on millions of documents: semantic search, topic classification, text similarity, and price recommendation.

Selected outcomes:

- Built a semantic backlink search system across millions of documents, reducing search latency from about 12 seconds to 3 seconds.
- Replaced mostly tag-based or random backlink discovery with similarity-based recommendations.
- Improved backlink suggestion relevance, contributing to positive user feedback and continued investment in the feature.
- Developed topic classification, text similarity, and price recommendation models for SEO workflows.
- Analyzed product usage and dissatisfaction signals to support SEO specialists and product decisions.

Technologies: Python, scikit-learn, fastText, word2vec, Gensim, Pandas, NumPy, ClickHouse, Flask, pytest, Jupyter, Power BI

Lead Ruby on Rails developer

4 years

OneRetarget.com

December 2016 - November 2020

<https://oneretarget.com>

A service to manage several ad platforms at one account

Led backend and full-stack development for an ad-tech product integrating multiple advertising platforms. Worked on technical decisions, API design, code review, testing, deployment, monitoring, and frontend modernization.

Selected outcomes:

- Adapted the API for both external users and frontend use, reducing duplicated integration logic.
- Built tools for advertising analytics, moderation-state synchronization, and campaign management.
- Added system problem-detection tooling to make operational issues easier to find and investigate.
- Implemented interactive UI features such as weekly schedules, bulk copying, and ad previews.
- Supported frontend migration from CoffeeScript/jQuery to ES6 and React.
- Helped unify account-wide UI layout and styling.

Technologies: Ruby on Rails, PostgreSQL, RSpec, Cucumber, Capybara, Docker, Jenkins, Capistrano, React, ES6, Redux

Ruby on Rails developer

8 months

Lakehouse

April 2016 - November 2016

<http://www.lakehouse.ru>

SaaS and in-house solutions

Supported and developed Ruby on Rails applications for real estate process management and professional qualification workflows.

- Supported and developed a real estate process-management project.
- Supported and developed a qualification/training project for beauty specialists.
- Worked on backend features, tests, deployment, and legacy frontend maintenance.

Technologies: Ruby on Rails 4, PostgreSQL, RSpec, Capybara, Capistrano, Linux, AngularJS, CoffeeScript

Self-education

3 months

February 2016 - April 2016

Built and rebuilt small web applications while transitioning into Ruby on Rails development.

- Rebuilt a student progress tracking application in Ruby on Rails.
- Developed a website for a trampoline center.

Projects:

<https://github.com/resivalex/student-progress-on-rails>

<https://github.com/resivalex/zaskok>

Technologies: Ruby on Rails, RSpec, Linux, Capistrano, AngularJS, CoffeeScript, PHP, HTML, SASS, MySQL, D3.js, SVG

Software developer

10 months

Nasca Ltd.

April 2015 - January 2016

<http://piratetrade.ru>

Exchange trading company

Worked on web development and trading-related software in a Scrum team. Built backend features, contributed to C++ networking components, wrote unit tests, and participated in code review.

- Developed a public website from scratch.
- Built parts of a multi-component network application for exchange-related workflows.
- Implemented a component responsible for interaction with the exchange.
- Contributed to Qt client design and implementation.
- Added unit tests using the Catch framework.
- Used design patterns and participated in code review.

Technologies: Ruby on Rails 4, Ruby Slim, JavaScript, CSS, SASS, Bootstrap, AJAX, C++, Boost Thread, WebSocket, JSON, Catch, JIRA, Confluence, Git

C++/Qt developer

10 months

Nasca Ltd.

December 2012 - September 2013

<http://piratetrade.ru>

Exchange trading company

Worked on trading-related desktop and backend tools, including trading engine improvements, log processing, exchange emulation, and automation utilities.

- Improved components of the trading engine.
- Built tools for exchange log parsing and analysis.
- Implemented features for transaction logging software.
- Developed an exchange emulator for testing and development.
- Added automated exchange log downloading and user email notifications.
- Used version control in a small engineering team.

Technologies: C++, C++ Standard Library, Qt, JavaScript, jQuery, HTML, SVN

Higher education

Pskov State University

2008 - 2013

Faculty of Informatics

Languages

Russian — Native

English — Upper Intermediate

Contacts

LinkedIn profile

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